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Adjoint Consistency of Spacetime Discontinuous Galerkin Discretizations for Incompressible Flows

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Adjoint-consistency of a spacetime discontinuous Galerkin discretization for the incompressible Navier-Stokes equations is considered and the spacetime continuous and discrete adjoint problems are derived. The spacetime discretization is used to compute primal and discrete adjoint solutions to problems involving an incompressible fluid moving through a channel with oscillating walls. We consider both the case of a prescribed inflow velocity that is parabolic and a pressure driven flow at Reynolds numbers of 100 and 250. In all cases, separation and reconnection is observed, with recirculation regions forming at one quarter wave period and repeating every half wave period. The adjoint solution is shown to be smooth at surfaces over which the cost function is evaluated, indicating the discretization is adjoint consistent.

I. Introduction

In recent years, there has been a resurgence of interest in high-order numerical methods for computational fluid dynamics.[1] The payoff is obvious: if the discretization error of a numerical method decreases at a rate of h^n for some characteristic discretization length scale h, then increasing the order, n, of the method would result in much more rapid convergence toward the true solution. For instance, halving the grid size would result in a reduction in error by a factor of 32 for a 5$^{\text{th}}$-order method compared to a factor of 4 for a 2$^{\text{nd}}$-order method. Historically, the tradeoffs to this improved performance were increased computational effort per degree of freedom and more cumbersome implementations, particularly on unstructured grids. The discontinuous Galerkin finite element method (DG-FEM)[2,3,4,5] provides a methodology to address these issues. A significant body of literature now exists describing the application of this method to the compressible Euler and Navier-Stokes[6,7,8,9] equations. It has also been applied to the incompressible Navier-Stokes equations[10,11,12,13] as well.

The discontinuous Galerkin (DG) method can be used to discretize both the temporal and spatial domain. The resulting discretization, known as spacetime DG provides a natural mechanism to handle grid motion, including the opening and closing of gaps, without distorting the mesh cells. It also allows for local refinement in both space and time. The adjoint consistent spacetime DG discretization recently described by the present authors is well suited for optimization problems.[14] Indeed, many of the numerical properties desired for industrial applications are inherent to spacetime DG.

Localization of the timestep is advantageous for problems that include advection and diffusion over a wide range of scales. In simulations of energy exchange, e.g., heat transfer, between a body and surrounding fluid, the wake is resolved, spatially, only to the extent needed to accurately capture shedding. However, temporally, the wake is resolved with the same accuracy as the region of interest near the body. Local timestepping with a spacetime discretization can lead to more accurate solutions by resolving the wake

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only to the extend needed in both time and space, shifting scarce computational resources from physically uninteresting regions to physically meaningful ones.

The spacetime discretization also provides natural way to handle moving bodies.[12,15,13] Traditional approaches to mesh movement such as Arbitrary Lagrangian- Eulerian (ALE) methods,[16,17] which provide excellent accuracy for moving bodies, may deteriorate when large deformations are induced by mesh motion, and also incur complexity due to the need to satisfy a Geometric Conservation Law (GCL).[18,19] For problems involving Fluid Structure Interactions (FSI), traditional methods fail when gaps between an object and a body degenerate and close. Spacetime discretizations offer an alternative by accurately tracking the body without distorting the mesh.

In this paper, we present a spacetime DG finite element method for the incompressible Navier-Stokes equations using a mixed formulation.[20] The continuous and discrete adjoint problems are derived, and the discretization of the cost function associated with pressure induced drag is shown to be adjoint consistent. Finally, primal and discrete adjoint solutions for problems involving moving bodies are presented. The adjoint solutions are shown to be smooth, indicating the discretization is adjoint consistent.

II. Methodology

II.A. The Incompressible Navier-Stokes Equations

In this work, we extend the adjoint-consistent spacetime, discontinuous Galerkin method presented by the present authors,[14] which itself is an extension of the adjoint-consistent discontinuous Galerkin method of Hartmann,[22,5] to a mixed formulation for incompressible flow problems. We consider a non-stationary spacetime domain $\mathcal{Q} \subset \mathbb{R}^{d+1}$ bounded by $\Sigma = \partial\mathcal{Q}$. In the case of stationary boundaries, the spacetime domain, $\mathcal{Q}$, can be expressed as the product of a bounded Lipschitz domain, $\Omega(t) \subset \mathbb{R}^d$, and positive interval $[0, T]$, i.e., $\mathcal{Q} \coloneqq \Omega \times T$, with $\Sigma \coloneqq \partial(\Omega \times T)$. For the incompressible Navier-Stokes equations the spacetime problem statement is

$$\nabla \cdot (\mathcal{F}^c(\mathbf{u}) - \mathcal{F}^v(\nabla\mathbf{u})) + \nabla_{\mathbf{x}} p = \mathbf{0} \qquad \text{in } \mathcal{Q}, \tag{1}$$

$$\nabla_{\mathbf{x}} \cdot \mathbf{u} = \mathbf{0} \qquad \text{in } \mathcal{Q}, \tag{2}$$

$$\mathbf{u} = G_D \qquad \text{in } \Sigma_D, \tag{3}$$

$$\mathbf{n} \cdot (\mathcal{F}^c(\mathbf{u}) - \mathcal{F}^v(\nabla\mathbf{u})) + \mathbf{n}_{\mathbf{x}} p = G_N \qquad \text{in } \Sigma_N. \tag{4}$$

$$\tag{5}$$

We use $\nabla_{\mathbf{x}}, \mathbf{n}_{\mathbf{x}}$ and ∇_t, n_t to denote the decomposition of differential operators and normal vectors into spatial and temporal components respectively. The velocity is denoted $\mathbf{u} = (u_1, \cdots, u_d)^\top$, while the pressure p is a Lagrange multiplier corresponding to the solution of the adjoint equation for the divergence constraint (2). The $d+1$ dimensional spacetime convective and viscous fluxes are given as

$$\boldsymbol{f}_i^c(\mathbf{u}) = \begin{pmatrix} u_1 u_i \\ \vdots \\ u_d u_i \\ u_i \end{pmatrix}, \quad \boldsymbol{f}_i^v(\nabla\mathbf{u}) = \begin{pmatrix} \tau_{1i} \\ \vdots \\ \tau_{di} \\ 0 \end{pmatrix},$$

for $i = 1, \ldots, d+1$ where $\tau = \nu\left(\nabla_{\mathbf{x}}\mathbf{u} + \nabla_{\mathbf{x}}\mathbf{u}^\top\right)$ is the stress tensor and ν is the dynamic viscosity. If the viscosity coefficient is constant, the divergence of the viscous flux reduces to a Laplacian, $\nabla \cdot \mathcal{F}^v = \nabla \cdot (\nu \nabla_{\mathbf{x}} \mathbf{u})$.

II.B. The continuous adjoint

Consider the cost functions for the total drag of an incompressible flow

$$J(\mathbf{u}, p) = \int_\Sigma j(\mathbf{u}, p)\, \mathrm{d}s = \int_{\Sigma_W} (p\mathbf{n}_{\mathbf{x}} - \tau\mathbf{n}_{\mathbf{x}}) \cdot \psi \, \mathrm{d}s \tag{6}$$

where $\psi = \left(2/\left(v_\infty^2 L_\infty\right)\right)(\cos\alpha, \sin\alpha)$, for angle of attack α, reference velocity v_∞, and reference length L_∞. The perturbation to the cost function, denoted δJ, about reference velocity $\mathbf{u}$ and pressure p, is defined

$$\delta J = J'[\mathbf{u}, p](\delta\mathbf{u}, \delta p) = J_{\boldsymbol{u}}[\boldsymbol{u}, p]\,\delta\boldsymbol{u} + J_p[\boldsymbol{u}, p]\,\delta p \tag{7}$$

then

$$\delta J = \int_{\Sigma_W} p\mathbf{n_x} \cdot \psi \delta p - \tau_{\nabla \mathbf{u}} \mathbf{n} \cdot \psi \nabla_{\mathbf{x}} \delta \mathbf{u}, \tag{8}$$

$$= (\delta p, \mathbf{n_x} \cdot \psi)_{\Sigma_W} - (\nabla_{\mathbf{x}} \delta \mathbf{u}, \tau_{\nabla \mathbf{u}} \mathbf{n} \cdot \psi)_{\Sigma_W}. \tag{9}$$

To derive the continuous adjoint, we first multiply (1, 2) by smooth functions $(\mathbf{u}^*, p^*)$ respectively, and linearize with respect to velocity, $\mathbf{u}$, and multiplier, p, to obtain: find $(\mathbf{u}^*, p^*) \in (V, Q)$ s.t.,

$$\left(\nabla \cdot \left(\mathcal{F}^c_{\mathbf{u}} \delta \mathbf{u} - \mathcal{F}^v_{\nabla \mathbf{u}} \nabla_{\mathbf{x}} \delta \mathbf{u}\right), \mathbf{u}^*\right)_{\mathcal{Q}} + \left(\nabla_{\mathbf{x}} \delta p, \mathbf{u}^*\right)_{\mathcal{Q}} + \left(\nabla_{\mathbf{x}} \cdot \delta \mathbf{u}, p^*\right)_{\mathcal{Q}} = \delta J, \tag{10}$$

where $\mathcal{F}^c_{\mathbf{u}}$ and $\mathcal{F}^v_{\nabla \mathbf{u}}$ denote the derivatives of $\mathcal{F}^c$ and $\mathcal{F}^v$ with respect to $\mathbf{u}$ and $\nabla_{\mathbf{x}} \mathbf{u}$, respectively. Integration by parts gives of (10) and rearranging terms gives

$$\begin{aligned} - \left(\left(\mathcal{F}^c_{\mathbf{u}} \delta \mathbf{u} - \mathcal{F}^v_{\nabla \mathbf{u}} \nabla_{\mathbf{x}} \delta \mathbf{u}\right), \nabla \mathbf{u}^*\right)_{\mathcal{Q}} - \left(\delta \mathbf{u}, \nabla_{\mathbf{x}} p^*\right)_{\mathcal{Q}} - \left(\delta p, \nabla_{\mathbf{x}} \cdot \mathbf{u}^*\right)_{\mathcal{Q}} \\ + \left(\mathbf{n} \cdot \left(\mathcal{F}^c_{\mathbf{u}} \delta \mathbf{u} - \mathcal{F}^v_{\nabla \mathbf{u}} \nabla_{\mathbf{x}} \delta \mathbf{u}\right), \mathbf{u}^*\right)_{\Sigma} + \left(\mathbf{n_x} \cdot \delta \mathbf{u}, p^*\right)_{\Sigma} + \left(\mathbf{n_x} \delta p, \mathbf{u}^*\right)_{\Sigma} = \delta J. \end{aligned} \tag{11}$$

A final application of integration by parts on (11) yields the variational formulation of the continuous adjoint problem for the incompressible Navier-Stokes equations on a spacetime domain, find $\mathbf{u}^*, p^*$ such that

$$\begin{aligned} - \left(\delta \mathbf{u}, \left(\mathcal{F}^c_{\mathbf{u}}\right)^\top \nabla \lambda + \nabla_{\mathbf{x}} \cdot \left(\left(\mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \nabla \mathbf{u}^*\right) + \nabla_{\mathbf{x}} p^*\right)_{\mathcal{Q}} - \left(\delta p, \nabla_{\mathbf{x}} \cdot \mathbf{u}^*\right)_{\mathcal{Q}} \\ + \left(\delta \mathbf{u}, \left(\mathbf{n} \cdot \mathcal{F}^c_{\mathbf{u}}\right)^\top \mathbf{u}^* + \mathbf{n_x} \cdot \left(\left(\mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \nabla \mathbf{u}^*\right) + \mathbf{n_x} p^*\right)_{\Sigma_N} \\ + \left(\delta p, \mathbf{n_x} \cdot \mathbf{u}^*\right)_{\Sigma_W} - \left(\nabla_{\mathbf{x}} \delta \mathbf{u}, \left(\mathbf{n} \cdot \mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \mathbf{u}^*\right)_{\Sigma_W} = \delta J. \end{aligned} \tag{12}$$

Specifically, for the cost function (6), solutions, $\mathbf{u}^*, p^*$, of the continuous adjoint equations satisfy

$$- \left(\left(\mathcal{F}^c_{\mathbf{u}}\right)^\top\right) \nabla \mathbf{u}^* - \nabla_{\mathbf{x}} \cdot \left(\left(\mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \nabla \mathbf{u}^*\right) - \nabla_{\mathbf{x}} p^* = \mathbf{0} \qquad \text{in } \mathcal{Q}, \tag{13}$$

$$-\nabla_{\mathbf{x}} \cdot \mathbf{u}^* = 0 \qquad \text{in } \mathcal{Q}, \tag{14}$$

with boundary conditions

$$\mathbf{u}^* = \mathbf{0} \qquad \text{in } \Sigma_D, \tag{15}$$

$$\left(\mathbf{n} \cdot \mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \cdot \mathbf{u}^* = \tau_{\nabla \mathbf{u}} \mathbf{n} \cdot \psi \qquad \text{in } \Sigma_W, \tag{16}$$

$$\left(\mathbf{n} \cdot \mathcal{F}^c_{\mathbf{u}}\right)^\top \mathbf{u}^* + \mathbf{n_x} \cdot \left(\left(\mathcal{F}^v_{\nabla \mathbf{u}}\right)^\top \nabla \mathbf{u}^*\right) + \mathbf{n_x} p^* = \mathbf{0} \qquad \text{in } \Sigma_N. \tag{17}$$

II.C. The discontinuous Galerkin discretization

The spacetime domain, $\mathcal{Q}$, is decomposed into non-overlapping finite elements, denoted as $\mathcal{T}_h$. The discrete spaces of mixed discontinuous polynomials is then defined as, for $p \in \mathbb{N}$,

$$\mathbf{V}^{p+1}_h := \left\{\mathbf{v}_h \in \left[L^2\left(\mathcal{Q}\right)\right]^d : \mathbf{v}_{h|\kappa} \in \left[\mathbb{P}_{p+1}(\kappa)\right]^d \forall \kappa \in \mathcal{T}_h\right\}, \tag{18}$$

$$Q^p_h := \left\{q_h \in L^2\left(\mathcal{Q}\right) : q_{h|\kappa} \in \mathbb{P}_p(\kappa) \forall \kappa \in \mathcal{T}_h\right\}, \tag{19}$$

$$\tag{20}$$

We define the jump and average operators for scalar and vector valued functions $(\mathbf{v}_h, q_h) \in \left(\mathbf{V}^{p+1}_h \times Q^p_h\right)$ as

$$\{\{q\}\} = \frac{1}{2}(q^+ + q^-) \text{ in } \Sigma_I, \qquad \{\{q\}\} = q^+ \text{ in } \Sigma,$$

$$[\![q]\!] = (q^+ \mathbf{n}^+ + q^- \mathbf{n}^-) \text{ in } \Sigma_I, \qquad [\![q]\!] = q^+ \mathbf{n}^+ \text{ in } \Sigma,$$

for scalar valued functions $q_h \in Q^p_h$,

$$\{\{\mathbf{v}\}\} = \frac{1}{2}(\mathbf{v}^+ + \mathbf{v}^-) \text{ in } \Sigma_I, \qquad \{\{\mathbf{v}\}\} = \mathbf{v}^+ \text{ in } \Sigma,$$

$$[\![\mathbf{v}]\!] = (\mathbf{v}^+ \cdot \mathbf{n}^+ + \mathbf{v}^- \cdot \mathbf{n}^-) \text{ in } \Sigma_I, \qquad [\![\mathbf{v}]\!] = \mathbf{v}^+ \cdot \mathbf{n}^+ \text{ in } \Sigma,$$

for vector valued functions $\mathbf{v}_h \in \mathbf{V}_h^{p+1}$.

Following Hartmann,[22,5] the nonlinear convection operator is discretized as,

$$-\int_{\mathcal{Q}} \mathcal{F}^c(\mathbf{u}_h) : \nabla \mathbf{v}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \hat{\mathbf{h}} \,\mathrm{d}s, \tag{21}$$

$$\tag{22}$$

where $\hat{\mathbf{h}}$ is a consistent and conservative numerical flux. The viscous operator is discretized with the symmetric interior penalty Galkerin (SIPG),

$$\int_{\mathcal{Q}} \left(\mathcal{F}^v\left(\nabla \mathbf{u}_h\right)\right) : \nabla \mathbf{v}_h \,\mathrm{d}\mathbf{x} - \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \left(\hat{\sigma_h} \mathbf{n}_{\mathbf{x}}\right) \cdot \mathbf{v}_h \,\mathrm{d}s + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \left(\hat{\mathbf{u}}_h - \mathbf{u}_h\right) \otimes \mathbf{n}_{\mathbf{x}} : \nabla \mathbf{v}_h \,\mathrm{d}s. \tag{23}$$

The pressure term of (1) and divergence constraint are discretized as

$$-\int_{\mathcal{Q}} p_h \nabla_{\mathbf{x}} \cdot \mathbf{v}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \hat{p}_h \mathbf{n}_{\mathbf{x}} \cdot \mathbf{v}_h \,\mathrm{d}s = \mathbf{0}, \tag{24}$$

$$-\int_{\mathcal{Q}} q_h \nabla_{\mathbf{x}} \cdot \mathbf{u}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} q_h \mathbf{n}_{\mathbf{x}} \cdot [\![\mathbf{u}_h]\!] \,\mathrm{d}s = \mathbf{0} \tag{25}$$

where the trace of $\hat{p} = \{\{p\}\}$ following Rivi‘ere.[3]

Following Hartmann and Leicht[22,5] and Rivi‘ere,[3] we arrive at the discretization of (1,2) in element based form, find $(\mathbf{v}_h, q_h) \in \left(\mathbf{V}_h^{p+1} \times Q_h^p\right)$ such that

$$\begin{aligned} N_h\left(\mathbf{u}_h, \mathbf{v}_h\right) = &\int_{\mathcal{Q}} \left(-\mathcal{F}^c(\mathbf{u}_h) + \mathcal{F}^v\left(\nabla \mathbf{u}_h\right)\right) : \nabla \mathbf{v}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \left(\hat{\mathbf{h}} - \hat{\sigma}_h \mathbf{n}\right) \cdot \mathbf{v}_h \,\mathrm{d}s \\ &+ \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \left(\hat{\mathbf{u}}_h - \mathbf{u}_h\right) \otimes \mathbf{n}_{\mathbf{x}} : \nabla_{\mathbf{x}} \mathbf{v}_h \,\mathrm{d}s - \int_{\mathcal{Q}} p_h \nabla_{\mathbf{x}} \cdot \mathbf{v}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} \hat{p}_h \mathbf{n}_{\mathbf{x}} \cdot \mathbf{v}_h \,\mathrm{d}s = \mathbf{0} \;\forall \mathbf{v}_h \in \mathbf{V}_h^{p+1}, \end{aligned} \tag{26}$$

$$B_h\left(q_h, \mathbf{v}_h\right) = -\int_{\mathcal{Q}} q_h \nabla_{\mathbf{x}} \cdot \mathbf{u}_h \,\mathrm{d}\mathbf{x} + \sum_{\kappa \in \mathcal{T}_h} \int_{\Sigma} q_h \mathbf{n}_{\mathbf{x}} \cdot [\![\mathbf{u}_{\mathbf{h}}]\!] \,\mathrm{d}s = 0 \;\forall q_h \in Q_h^p. \tag{27}$$

The numerical fluxes, $\hat{\mathbf{u}}_h$ and $\hat{\sigma}_h$, are defined following[5] as

$$\hat{\mathbf{u}}_h = \{\{\mathbf{u}_h\}\}, \qquad \hat{\sigma}_h = \left(\{\{\mathcal{F}^v_{\nabla \mathbf{u}} \nabla_{\mathbf{x}} \mathbf{u}_h\}\} - C_{ip} \frac{p^2}{h_e} \{\{\mathcal{F}^v_{\nabla \mathbf{u}}\}\} [\![\mathbf{u}_{\mathbf{h}}]\!]\right) \text{ in } \Sigma, \tag{28}$$

where h_e is an appropriate measure for the length scale, p is the polynomial degree, and C_{ip} is a constant choosen such that stability is guaranteed. The numerical flux associated with convection, $\hat{\mathbf{h}}$, is composed of a spatial and temporal fluxes:

$$\hat{\mathbf{h}} = \hat{\mathbf{h}}_x + \hat{\mathbf{h}}_t \tag{29}$$

where $\hat{\mathbf{h}}_x$ is a consistent and conservative numerical flux for the spatial components of the spacetime flux, and $\hat{\mathbf{h}}_t$ is defined as the upwind flux of the temporal component of the spacetime flux.

II.D. The Discrete Adjoint

The discrete adjoint for the mixed discontinuous Galerkin discretization (26,27) of the incompressible Navier-Stokes equation (1,2) for a consistent discretization of the cost function (6) given as

$$J_h\left(\mathbf{u}_h, p_h\right) = \int_{\Sigma_W} \left(\hat{p}_{\Sigma,h} \mathbf{n}_{\mathbf{x}} - \hat{\sigma}_{\Sigma,h} \mathbf{n}_{\mathbf{x}}\right). \tag{30}$$

where $\hat{p}_{\Sigma,h}$ and $\hat{\sigma}_{\Sigma,h}$ are consistent discretiztions of the pressure and diffusive numerical flux at the boundary. Then Frèchet derivative of $J_h(\mathbf{u}_h, p_h)$ in (30) with respect to $(\mathbf{u}_h, p_h)$ in the direction $(\delta\mathbf{u}_h, \delta p_h)$

$$J_h'[\mathbf{u}_h, p_h](\delta\mathbf{u}_h, \delta p_h) = \int_{\Sigma_W} \mathbf{n_x} \cdot \psi \delta p_h - (\sigma_{\nabla \mathbf{u_h}} \mathbf{n}) \cdot \psi \nabla_{\mathbf{x}} \delta \mathbf{u_h}. \tag{31}$$

Denoting Frèchet derivative of $N_h(\mathbf{u}_h, p_h)$ in (26) with respect to $(\mathbf{u}_h, p_h)$ in the direction $(\delta\mathbf{u}_h, \delta p_h)$ as $N_h'[\mathbf{u}_h, p_h](\delta\mathbf{u}_h, \delta p_h, \mathbf{u}^*, p^*) = \delta N_h$ and the Frèchet derivative of $B_h(\mathbf{u}_h)$ in (27) with respect to $(\mathbf{u}_h)$ in the direction $(\delta\mathbf{u}_h)$ as $B_h'[\mathbf{u}_h](\delta\mathbf{u}_h, \mathbf{u}^*) = \delta B_h$ we have

$$\begin{aligned} \delta N_h = \int_{\mathcal{Q}} \left(-\mathcal{F}^c_{\mathbf{u}}[\mathbf{u}_h]\delta\mathbf{u} + \mathcal{F}^v_{\nabla\mathbf{u}}[\mathbf{u}_h]\nabla_{\mathbf{x}}\delta\mathbf{u}\right) : \nabla\mathbf{u}_h^* \,\mathrm{d}\mathbf{x} + \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\left(\hat{\mathbf{h}}_{\mathbf{u}^+}\delta\mathbf{u}_h^+ - \hat{\sigma}_{\nabla_{\mathbf{x}}\mathbf{u}^+}\nabla_{\mathbf{x}}\delta\mathbf{u}_h^+\mathbf{n_x}\right)\cdot\left(\mathbf{u}_h^{*+} - \mathbf{u}_h^{*-}\right)\mathrm{d}s \\ + \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\left(\hat{\mathbf{u}}_{\mathbf{u}^+}\delta\mathbf{u}_h^+\right)\otimes\mathbf{n_x} : \left(\nabla_{\mathbf{x}}\mathbf{u}_h^{*+} - \nabla_{\mathbf{x}}\mathbf{u}_h^{*-}\right)\mathrm{d}s - \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\delta\mathbf{u}_h^+\otimes\mathbf{n_x} : \nabla_{\mathbf{x}}\mathbf{u}_h^*\,\mathrm{d}s \\ + \int_{\Sigma}\left(\hat{\mathbf{u}}_{\Sigma,\mathbf{u}}\delta\mathbf{u}_h\right)\otimes\mathbf{n_x} : \nabla_{\mathbf{x}}\mathbf{u}_h^*\,\mathrm{d}s + \int_{\Sigma}\left(\hat{\mathbf{h}}_{\Sigma,\mathbf{u}} - \hat{\sigma}_{\Sigma,\nabla\mathbf{u}}\mathbf{n_x}\right)\cdot\mathbf{u}_h^*\,\mathrm{d}s \\ + \int_{\mathcal{Q}}\nabla\delta p_h\mathbf{u}_h^*\,\mathrm{d}\mathbf{x} - \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\hat{\delta p_h}\mathbf{n_x}\cdot\mathbf{u}_h^*\,\mathrm{d}s = \mathbf{0}, \end{aligned} \tag{32}$$

$$\delta B_h = -\int_{\mathcal{Q}} p_h^*\nabla_{\mathbf{x}}\cdot\delta\mathbf{u}_h\,\mathrm{d}\mathbf{x} + \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma} p_h^*\mathbf{n_x}\cdot[\![\delta\mathbf{u_h}]\!]\,\mathrm{d}s = 0. \tag{33}$$

Finally, we note (6) is a consistent discretization of the cost function since the cost function and discretization are equal when evaluated for the exact solution p, i.e., $\hat{p}_\Gamma = p$ on Γ, thus $p\mathbf{n_x}\cdot\psi = p\mathbf{n_x}\cdot\psi$.

II.E. Simulation Method Details

The bilinear form of (26,27) is

$$N_h(\mathbf{u}_h, \mathbf{v}_h) = \int_{\mathcal{Q}}\left(-\mathcal{F}^c(\mathbf{u}_h) + \mathcal{F}^v(\nabla\mathbf{u}_h)\right) : \nabla\mathbf{v}_h\,\mathrm{d}\mathbf{x} - \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\left(\hat{\mathbf{h}} + \hat{\sigma}_h\mathbf{n}\right)\cdot\mathbf{v}_h\,\mathrm{d}s \tag{34}$$

$$+ \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma}\left(\hat{\mathbf{u}}_h - \mathbf{u}_h\right)\otimes\mathbf{n_x} : \nabla_{\mathbf{x}}\mathbf{v}_h\,\mathrm{d}s, \tag{35}$$

$$B_h(q_h, \mathbf{v}_h) = -\int_{\mathcal{Q}} q_h\nabla_{\mathbf{x}}\cdot\mathbf{v}_h\,\mathrm{d}\mathbf{x} + \sum_{\kappa\in\mathcal{T}_h}\int_{\Sigma} q_h[\![\mathbf{v}_h]\!]\,\mathrm{d}s, \tag{36}$$

In matrix form,

$$\begin{pmatrix} \mathbf{N}_h & \mathbf{B}_h^\top \\ \mathbf{B}_h & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{U}_h \\ \mathbf{P}_h \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix},$$

Newton's method is applied to the linearized system given as

$$\begin{pmatrix} \mathbf{A}_h & \mathbf{B}_h^\top \\ \mathbf{B}_h & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{U}_h \\ \mathbf{P}_h \end{pmatrix} = \begin{pmatrix} -\mathbf{f_u} \\ -\mathbf{f}_p \end{pmatrix},$$

where A is the linearization of N, defined in (34), and $\mathbf{f_u}$ and $\mathbf{f}_p$ are the nonlinear residuals. We have negated the divergence constraint (2) for symmetry of the diagonal terms.

III. Results:

We present the primal and corresponding discrete adjoint solutions for problems involving moving boundaries. Specifically, results are presented for problems involving incompressible flows through a channel with oscillating walls. The walls of the channel are initially at rest and the solution corresponds to a fully developed Poiseuille flow, where the velocity has a parabolic profile and pressure has a linear gradient. For $t > 0$, the position of the of the channel walls, (x, y, t), is parameterized by

$$s(x,t) = A\sin(m\pi x)\sin(n\pi t) + H \tag{37}$$

and

$$y = \begin{cases} s(x,t) & \text{if } y \geq 0 \\ -s(x,t) & \text{if } y < 0 \end{cases}$$

for $x, y, t \in [0, L] \times [-H, H] \times [0, T]$, where L is the length of the domain, H is one half the channel width, T is the final time, A is the amplitude of the wave, m and n, are the spatially and temporal wave numbers respectively. In the solutions given below, unless otherwise stated, were computed with $L = 1$, $H = T = 0.5$, $A = 0.2$, $m = 4$, and $n = 2$.

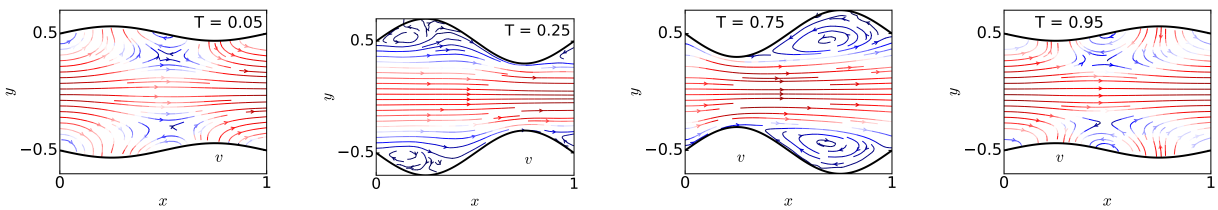

Figure 1. Streamlines throughout one wave period of the flow solution through an oscillating channel where $Re = 100$ at the parabolic inflow $x = 0$.

On the channel walls, no-slip boundary conditions are imposed, where the flow has a zero velocity in the wall's frame of reference. On the planes $x = L$ and $t = T$, the standard outflow condition for incompressible flows is applied, i.e., $G_N = 0$. On the plane $t = 0$, a parabolic velocity of the form

$$u_1 = -\frac{3}{2}H^2\left(\left(\frac{y}{H}\right)^2 - 1\right) \tag{38}$$

where the mean flow is assumed to be one. On the plane $x = 0$, we first consider the case of a parabolic inflow given by (38) so that it is equivalent with the condition on the temporal inflow plane at $t = 0$. We then consider the case of a pressure driven flow, where the pressure is chosen to be consistent with the solution at $t = 0$. For Reynolds numbers 100 and 250, G_N on the plane $x = 0$, is given as 0.12 and 0.024 respectively.

The motion of the channel walls is perpendicular to the direction of the initial parabolic flow. This cyclic motion causes repeated separation and reconnection of the flow throughout the cycle of the wave with regions of recirculation forming every $nt/4$ and $3nt/4$ as shown in Figure 1 at $t = 0.25$ and $t = 0.75$ respectively. It is unclear if the increased size of the recirculation region at $t = 0.75$, as compared to $t = 0.25$, is due to the greater core velocity at the later time or if it is an effect of boundaries.

Below we present primal and discrete adjoint solutions to flows through an oscillating channel for cost functions of the form

$$J_h(\mathbf{u}_h) = \frac{1}{2}\int_{\Sigma_N} \left(u - \hat{u}_{\Sigma,h}\right)^2 d\mathbf{s}. \tag{39}$$

where $u = (u_1, 0)$ for u_1 given by (38) and Σ_N corresponds to the temporal outflow plane, i.e., $t = T$.

The primal solutions at $R_e = 250$ for one half wave cycle are shown at times $t = 0.15, 0.25, 0.45$ in the top row of Figure 2. At $t = 0.25$, reattachment occurs farther along the channel wall than for the case of $R_e = 100$, while most other flow features are very similar to the $R_e = 100$ solution. The corresponding discrete adjoint solutions for the cost function (39), evaluated over the surface $t = T$, are shown is in the bottom row of 2.

The primal solution for a pressure driven flow and the corresponding discrete adjont solutions for the cost function (39), evaluated over the surface $t = T$ at times $t = 0.05, 0.25, 0.45$, are shown in Figure 3. As with the case of a parabolic inflow, streamlines of the adjoint solution are smooth, indicating the discretization is indeed adjoint consistent.

The solution of the discrete adjoint corresponding to the cost function (39) evaluated over the spatial outflow, i.e., $x = L$ during the first half wave cycle is shown in the bottom row of Figure 3 at times $t =$

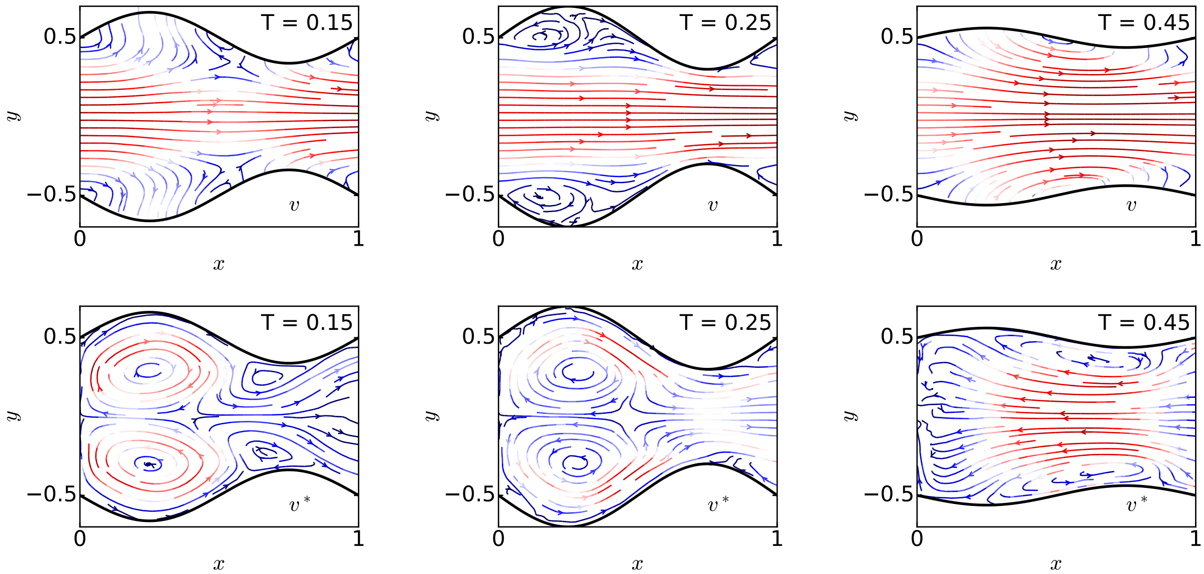

Figure 2. The primal (top) and discrete adjoint (bottom) solutions of flow through an oscillating channel where $Re = 250$ at the parabolic inflow $x = 0$ and the cost function $J_{h\Sigma_N}$ is evaluated over the temporal outflow plane $t = T$.

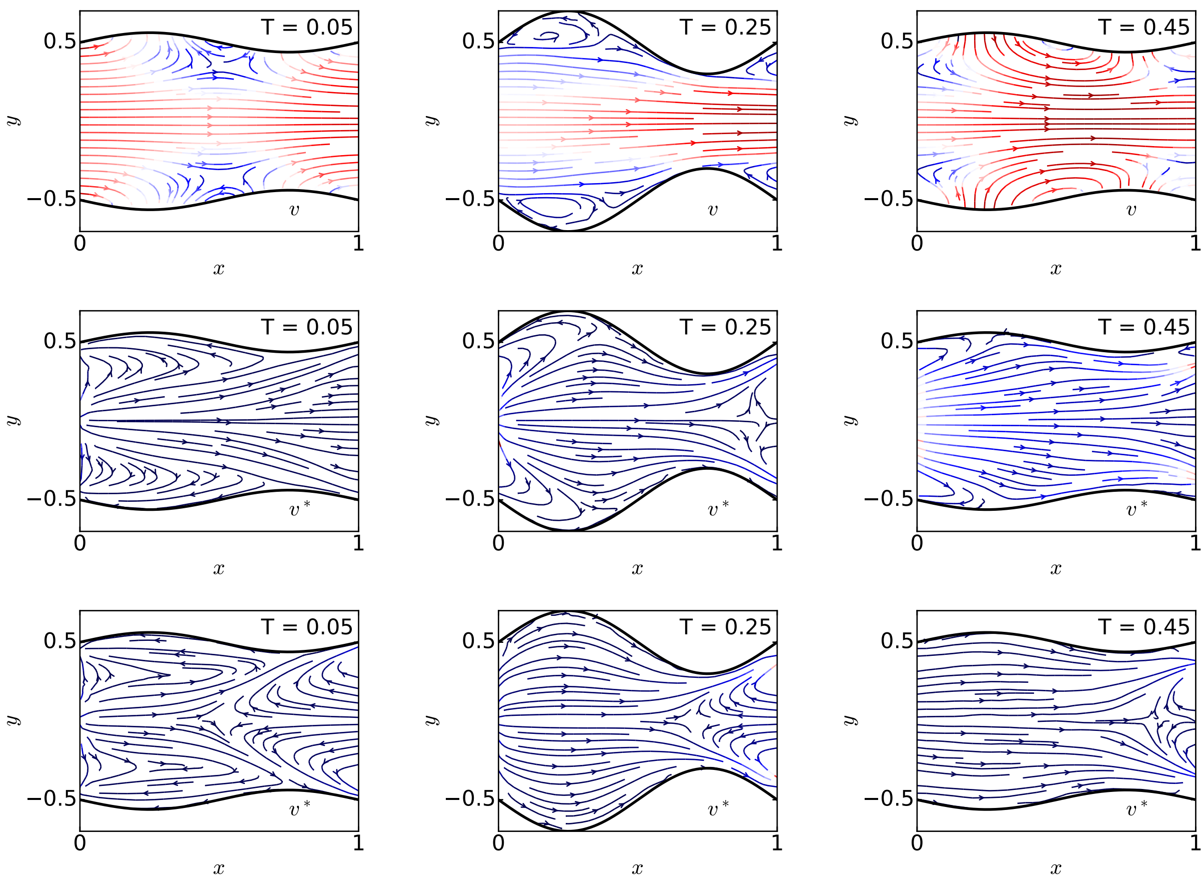

Figure 3. The primal (top) and discrete adjoint solutions for a pressure driven flow, initially at $Re = 250$, through an oscillating channel where the cost function $J_{h\Sigma_N}$ is evaluated over the temporal outflow plane $t = T$ (middle) and the spatial outflow plane $x = L$ (bottom).

$0.05, 0.25, 0.45$. Again, smoothness of the adjoint solution indicates adjoint consistency of the discretization in combination with the cost function (39).

IV. Conclusions

The continuous adjoint for the incompressible Navier-Stokes equations, as well as the adjoint boundary conditions for cost functions associated with optimization were presented. The discrete adjoint for spacetime discontinous Galerkin discretization of the incompressible Navier-Stokes equations was derived and used to compute adjoint solutions for problems involving moving boundaries. Discrete adjoint solutions for both prescribed parabolic inflow and pressure driven flow for a cost function defined over the plane corresponding to one half wave cycle were given. In the case of a pressure driven flow, the solution to the discrete adjoint problem was computed for the cost function defined over the spatial outflow for the entirety of one half wave cycle. All discrete adjoint solutions were shown to be smooth, indicating the discretization is adjoint consistent.

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References

[1]Wang, Z., Fidkowski, K., Abgrall, R., Bassi, F., Caraeni, D., Cary, A., Deconinck, H., Hartmann, R., Hillewaert, K., Huynh, H., et al., "High-order CFD methods: current status and perspective," *International Journal for Numerical Methods in Fluids*, Vol. 72, No. 8, 2013, pp. 811–845.

[2]Arnold, D. N., Brezzi, F., Cockburn, B., and Marini, L. D., "Unified analysis of discontinuous Galerkin methods for elliptic problems," *SIAM journal on numerical analysis*, Vol. 39, No. 5, 2002, pp. 1749–1779.

[3]Rivière, B., *Discontinuous Galerkin Methods For Solving Elliptic And Parabolic Equations: Theory and Implementation*, Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 2008.

[4]Di Pietro, D. A. and Ern, A., *Mathematical aspects of discontinuous Galerkin methods*, Vol. 69, Springer Science & Business Media, 2011.

[5]Hartmann, R. and Leicht, T., "Higher order and adaptive DG methods for compressible flows," *VKI LS 2014-03: 37th Advanced VKI CFD Lecture Series: Recent developments in higher order methods and industrial application in aeronautics, Dec. 9-12, 2013*, edited by H. Deconinck, Von Karman Institute for Fluid Dynamics, Rhode Saint Genèse, Belgium, 2014.

[6]Baumann, C. E. and Oden, J. T., "A discontinuous hp finite element method for the solution of the Euler equations of gas dynamics," *International Conference on Finite Elements in Fluids, 10 th, Tucson, AZ*, 1998, pp. 437–443.

[7]Fidkowski, K. J., Oliver, T. A., Lu, J., and Darmofal, D. L., "p-Multigrid solution of high-order discontinuous Galerkin discretizations of the compressible Navier–Stokes equations," *Journal of Computational Physics*, Vol. 207, No. 1, 2005, pp. 92–113.

[8]Luo, H., Baum, J. D., and Löhner, R., "A p-multigrid discontinuous Galerkin method for the Euler equations on unstructured grids," *Journal of Computational Physics*, Vol. 211, No. 2, 2006, pp. 767–783.

[9]Hartmann, R. and Houston, P., "An optimal order interior penalty discontinuous Galerkin discretization of the compressible Navier–Stokes equations," *Journal of Computational Physics*, Vol. 227, No. 22, 2008, pp. 9670–9685.

[10]Liu, J.-G. and Shu, C.-W., "A high-order discontinuous Galerkin method for 2D incompressible flows," *Journal of Computational Physics*, Vol. 160, No. 2, 2000, pp. 577–596.

[11]Nguyen, N. C., Peraire, J., and Cockburn, B., "An implicit high-order hybridizable discontinuous Galerkin method for the incompressible Navier–Stokes equations," *Journal of Computational Physics*, Vol. 230, No. 4, 2011, pp. 1147–1170.

[12]Rhebergen, S. and Cockburn, B., "A space–time hybridizable discontinuous Galerkin method for incompressible flows on deforming domains," *Journal of Computational Physics*, Vol. 231, No. 11, 2012, pp. 4185–4204.

[13]Rhebergen, S., Cockburn, B., and Van Der Vegt, J. J. W., "A Space-time Discontinuous Galerkin Method for the Incompressible Navier-Stokes Equations," *J. Comput. Phys.*, Vol. 233, Jan. 2013, pp. 339–358.

[14]Corrigan, A., Kessler, D., and Kercher, A., "An Adjoint-Consistent Space-Time Discontinuous Galerkin Method for Discontinuous Solutions," In Preparation.

[15]Neumüller, M., *Space-Time Methods: Fast Solvers and Applications*, Ph.D. thesis, Graz University of Technology, June 2013.

[16]Persson, P.-O., Bonet, J., and Peraire, J., "Discontinuous Galerkin solution of the Navier–Stokes equations on deformable domains," *Computer Methods in Applied Mechanics and Engineering*, Vol. 198, No. 17, 2009, pp. 1585–1595.

[17]Kast, S. M. and Fidkowski, K. J., "Output-based mesh adaptation for high order Navier–Stokes simulations on deformable domains," *Journal of Computational Physics*, Vol. 252, 2013, pp. 468–494.

[18]Lesoinne, M. and Farhat, C., "Geometric conservation laws for flow problems with moving boundaries and deformable

meshes, and their impact on aeroelastic computations," *Computer methods in applied mechanics and engineering*, Vol. 134, No. 1, 1996, pp. 71–90.

[19]Luo, H., Baum, J. D., and Löhner, R., "On the computation of multi-material flows using ALE formulation," *Journal of Computational Physics*, Vol. 194, No. 1, 2004, pp. 304–328.

[20]Brezzi, F. and Fortin, M., *Mixed and hybrid finite element methods*, Vol. 15, Springer Science & Business Media, 2012.

[21]Ramamurti, R. and L ohner, R., "Evaluation of an Incompressible Flow Solver Based on Simple Elements," *Advances in Finite Element Analysis in Fluid Dynamics*, Vol. 137, 1992, pp. 33–42.

[22]Hartmann, R., "Adjoint consistency analysis of discontinuous Galerkin discretizations," *SIAM Journal on Numerical Analysis*, Vol. 45, No. 6, 2007, pp. 2671–2696.

[23]Paige, C. C. and Saunders, M. A., "LSQR: An algorithm for sparse linear equations and sparse least squares," *ACM Transactions on Mathematical Software*, Vol. 8, No. 1, 1982, pp. 43–77.

[24]Paige, C. C. and Saunders, M. A., "Algorithm 583; LSQR: Sparse linear equations and least-squares problems," *ACM Transactions on Mathematical Software*, Vol. 8, No. 2, 1982, pp. 195–209.